

ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	94942
ToLID	qmAusTorr9
Species	Austropotamobius torrentium
Class	Malacostraca
Order	Decapoda

Genome Traits	Expected	Observed
Haploid size (bp)	10,622,541,117	12,403,682,275
Haploid Number	58 (source: ancestor)	90
Ploidy	2 (source: ancestor)	2
Sample Sex	unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 6.8.Q60

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed Haploid Number is different from Expected
- . Assembly length loss > 3% for pri
- . More than 1000 gaps/Gbp for pri

Curator notes

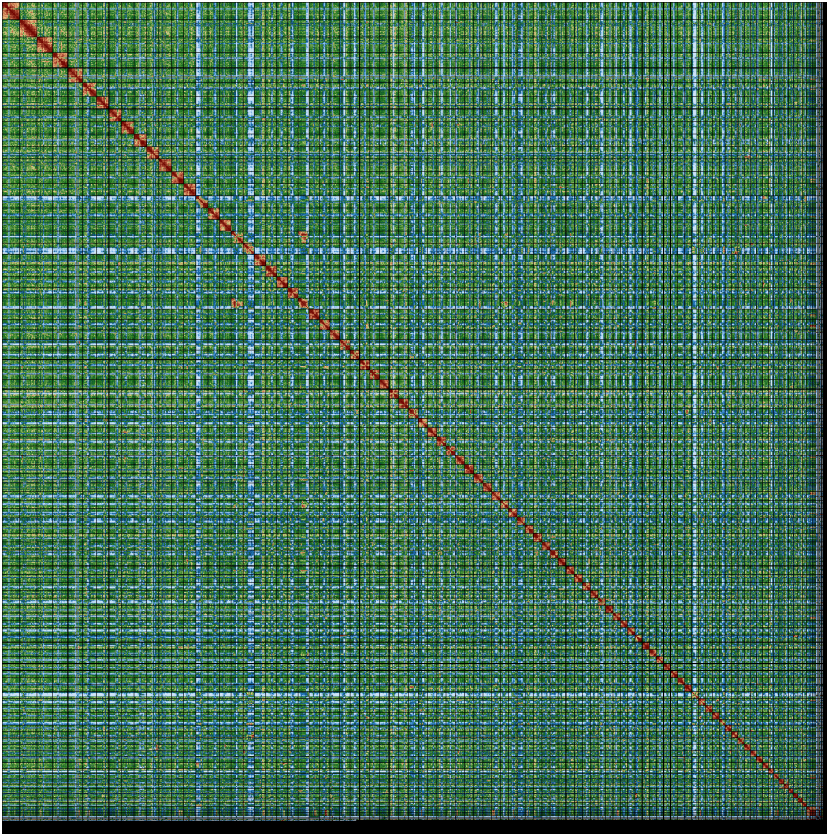
. Interventions/Gb: 36
. Contamination notes: "FCS-GX detected several bacteria and proteobacteria hits. 430 contigs (43Mb) were removed. Mitochondrial hits were removed"
. Other observations: "PacBio reads were assembled with hifiasm (HiC-mode, primary collapsed assembly selected). FCS-GX detected several contaminations. Purge_dups was run in default mode but removed more than 25% of content - primarily due to REPEAT-tagged sequences. Therefore the purging was done manually by filtering contigs based on mitohifi, btk, purge_dups and BUSCO results. Tiara was not used at this point, as the classification failed for the crayfish. Yaws was used in default mode to scaffold the contigs. Manual curation was done with PretextView. In a first round the scaffolds were roughly assigned to chromosomes and the remaining duplications were removed (~800Mb). The HiC Map was recreated for a second curation round. I ended up with 90 chromosomes, but I am not sure if that is correct. The genome is very repetitive and also shows several off-diagonal signals. Most of them repeat-induced. SUPER_18 and SUPER_24 show a strong signal. According to the literature Austropotamobius torrentium should have 88 chromosomes. I am open for any suggestions."

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	13,090,152,542	12,403,682,275
GC %	45.29	45.48
Gaps/Gbp	1,028.79	1,078.47
Total gap bp	1,346,700	1,365,000
Scaffolds	2,478	1,583
Scaffold N50	140,652,335	139,418,510
Scaffold L50	33	36
Scaffold L90	95	79
Contigs	15,945	14,960
Contig N50	2,245,118	2,164,511
Contig L50	1,353	1,313
Contig L90	6,847	6,625
QV	60.5143	60.4073
Kmer compl.	91.6926	91.0933
BUSCO sing.	95.3%	96.0%
BUSCO dupl.	3.1%	2.2%
BUSCO frag.	1.1%	1.0%
BUSCO miss.	0.5%	0.7%

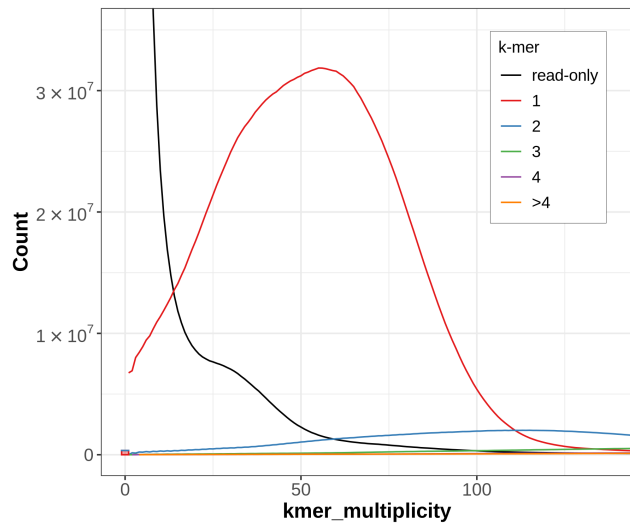
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: arthropoda_odb12 (genomes:76, BUSCOs:1667)

HiC contact map of curated assembly

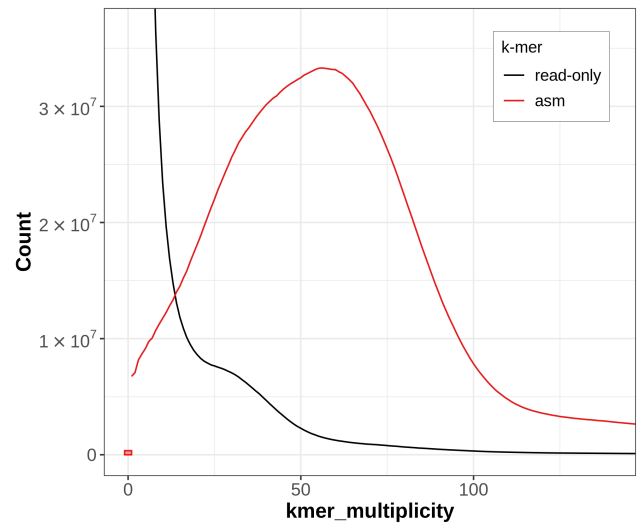


pri [\[LINK\]](#)

K-mer spectra of curated assembly

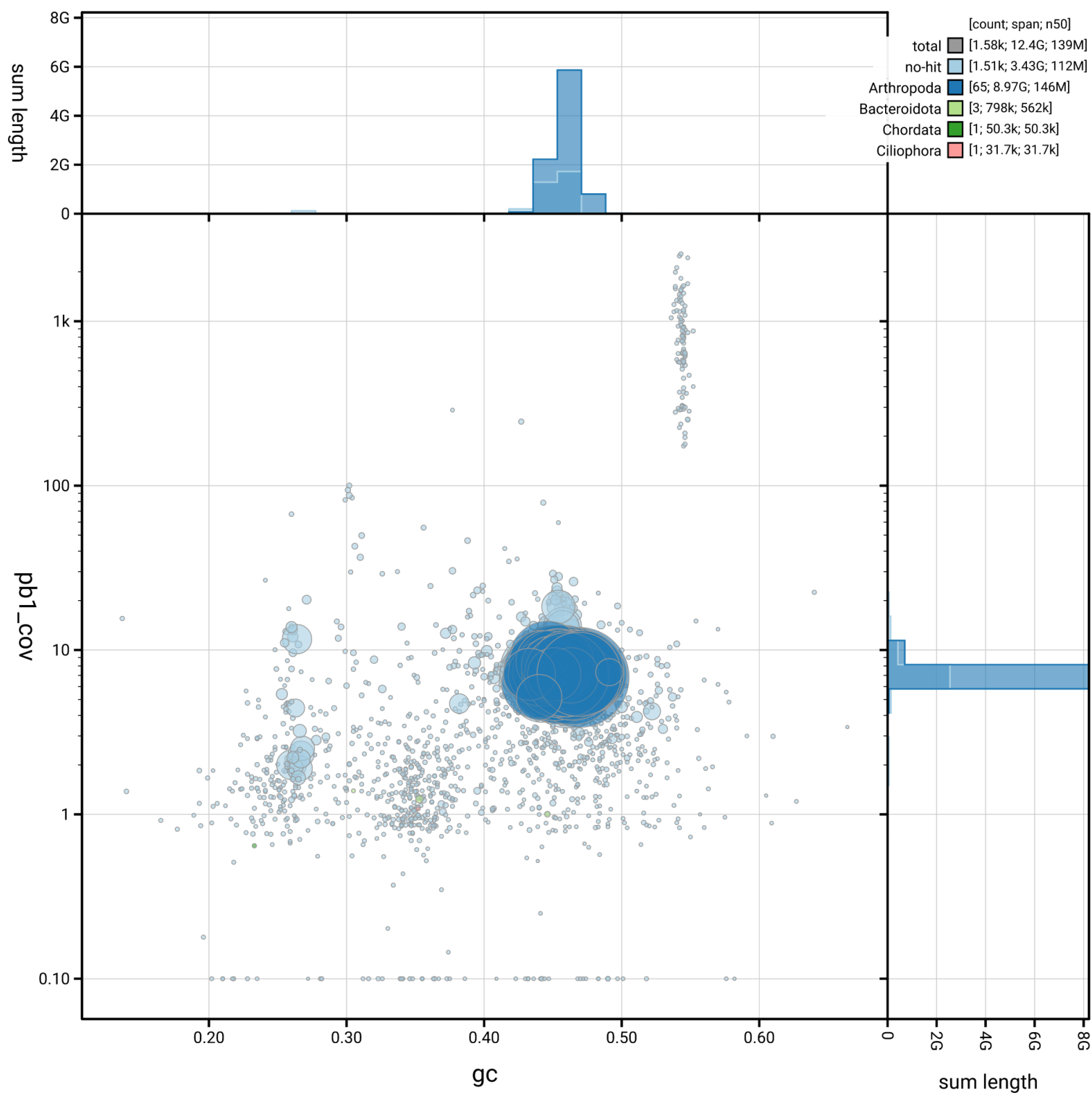


Distribution of k-mer counts per copy numbers found in asm



Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



pri. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	HiC
Coverage	60x	28x

Assembly pipeline

- **Hifiasm**
 - |_ *ver*: 0.25.0-r726
 - |_ *key param*: l2
- **purge_dups**
 - |_ *ver*: 1.2.6
 - |_ *key param*: NA

Curation pipeline

- **GRIT_Rapid**
 - |_ *ver*: 1a3d79a8
 - |_ *key param*: NA

Submitter: Martin Pippel

Affiliation: SciLifeLab

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